WEEK 8 PROJECT REPORT

STUDENT PERFORMANCE ANALYTICS DASHBOARD USING POWER BI

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1. INTRODUCTION

The Student Performance Analytics Dashboard is developed using Microsoft Power BI to analyse students' academic performance. The dashboard presents student scores, subject-wise averages, performance grades, pass/fail status, test preparation, and demographic information through interactive visualizations.

The main purpose of this project is to understand student performance patterns and present the findings in a clear and interactive format.

2. OBJECTIVES

- To analyse students' academic performance using Power BI.
- To compare average scores across Mathematics, Reading, and Writing.
- To analyse performance based on test preparation.
- To examine grade distribution and pass/fail status.
- To explore demographic factors such as parental education, gender, and race/ethnicity.
- To create an interactive dashboard with slicers and drill-through functionality.
- To identify meaningful, evidence-based insights from the dataset.

3. TOOLS AND TECHNOLOGIES USED

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Data visualization and interactive reporting
- Student Performance Dataset

4. DASHBOARD DEVELOPMENT

4.1 Data Preparation

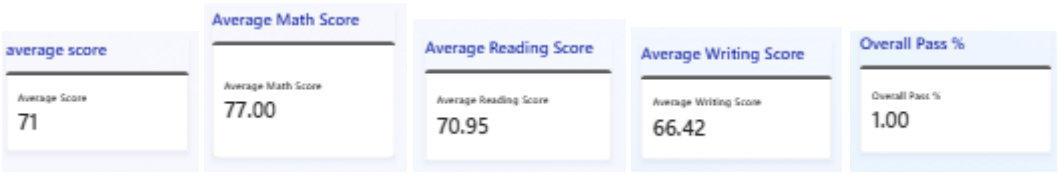
The student performance dataset was imported into Power BI. The data was inspected, cleaned, and transformed using Power Query Editor. Appropriate data types and columns were checked to prepare the dataset for analysis.

4.2 KPI Cards

KPI cards were created to display important performance indicators, including:

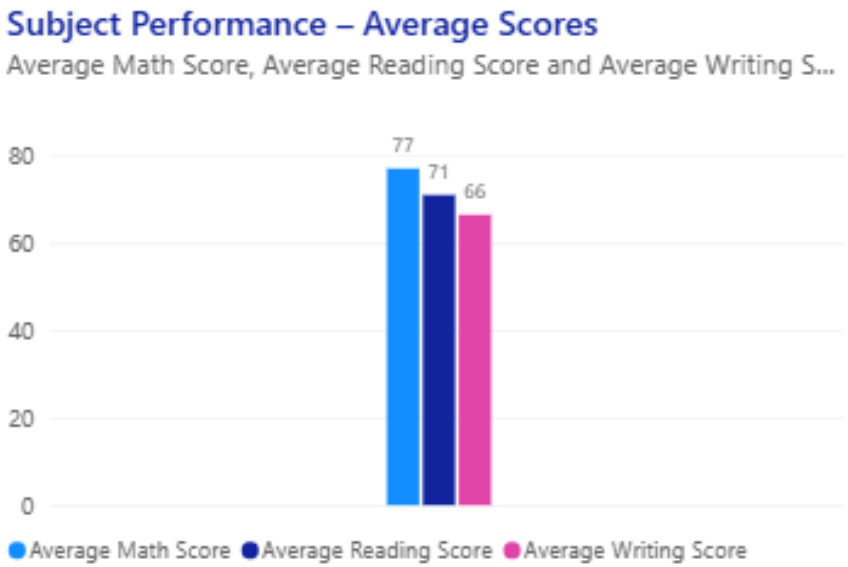
- Overall Average Score
- Average Mathematics Score
- Average Reading Score
- Average Writing Score
- Overall Pass Percentage

These cards provide a quick summary of student performance.



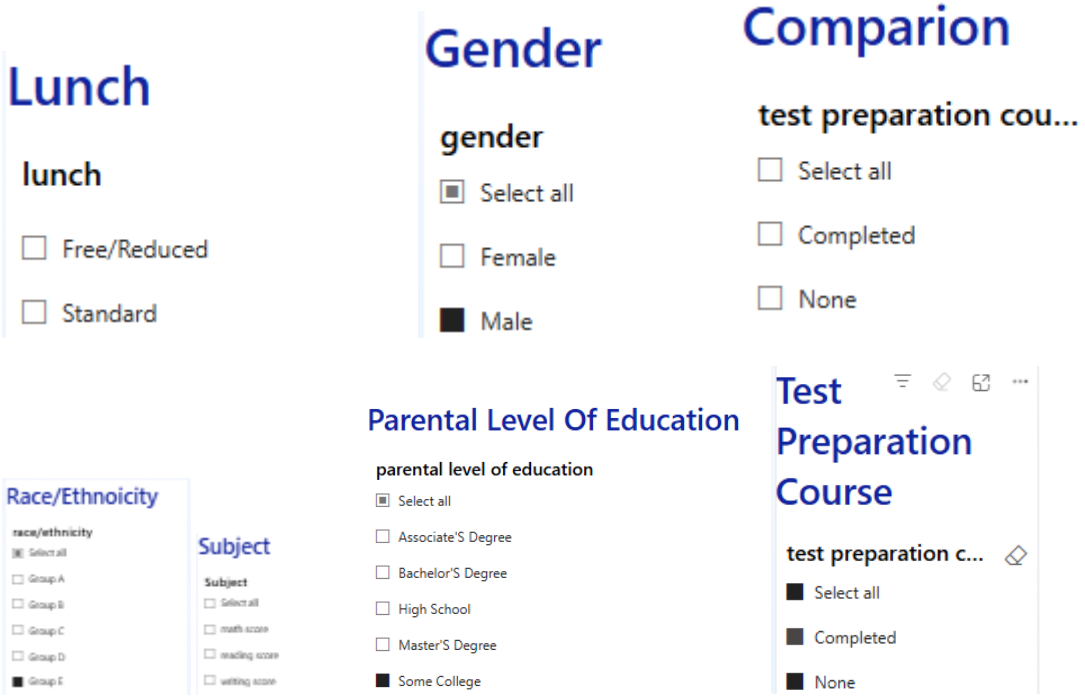
4.3 Subject Performance Analysis

A clustered column chart was created to compare average scores across Mathematics, Reading, and Writing. This visualization helps identify differences in performance among the three subjects.



4.4 Demographic Analysis

Visualizations and slicers were used to explore student performance across parental education, lunch type, gender, and race/ethnicity categories.



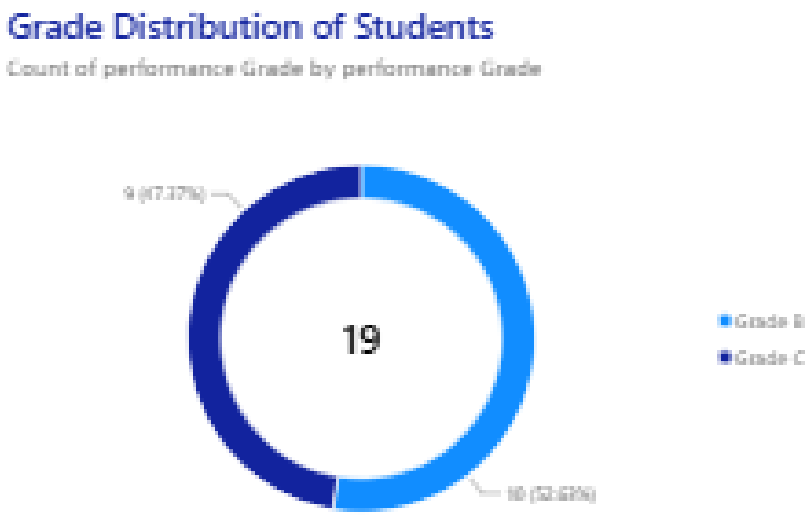
4.5 Test Preparation Analysis

A chart was created to compare the average scores of students who completed test preparation with those who did not complete it.



4.6 Grade Distribution

A chart was created to display the distribution of students across performance grade categories. This helps summarize the grades represented in the selected dashboard view.



4.7 Academic Profile – Drill-Through Page

A separate page named Academic Profile was created with a Student ID drill-through field. The page includes subject performance, Average Score, Performance Grade, Pass/Fail Status, and Test Preparation analysis.

A Back button was added and tested to return to the main dashboard.



5. DASHBOARD FORMATTING

The dashboard was formatted to improve readability and presentation. A consistent theme was maintained, meaningful visual titles were used, and visuals were aligned properly. KPI cards and charts were arranged to establish a clear visual hierarchy. Unnecessary visuals and excessive colours were avoided.

6. ANALYSIS AND INTERPRETATION

Insight 1: Subject Performance

Observation: Average scores differ across Mathematics, Reading, and Writing.

Evidence: Mathematics average score is 77.00, Reading average score is 70.95, and Writing average score is 66.42.

Interpretation: Mathematics has the highest average score, while Writing has the lowest average score among the three subjects in the displayed dashboard.

Insight 2: Test Preparation

Observation: Students who completed test preparation have a higher displayed average score than students who did not.

Evidence: Completed test preparation – 75; No test preparation – 70.

Interpretation: Test preparation completion is associated with a higher observed average score in this dataset. This comparison does not establish a causal relationship.

Insight 3: Grade Distribution

Observation: Students are distributed across different performance grades.

Evidence: In the displayed chart, Grade B has 10 students (52.63%), while Grade C has 9 students (47.37%).

Interpretation: Grade B represents a slightly larger share than Grade C in the current dashboard view. The counts may change when filters or slicers are changed.

Insight 4: Overall Student Performance

Observation: The dashboard provides an overall summary of student performance.

Evidence: The Overall Average Score card displays 71.

Interpretation: The overall average score provides a general summary of performance across the dataset and can be considered alongside subject-wise scores.

Insight 5: Parental Education

Observation: Student scores can vary across parental education categories.

Evidence: The displayed Average Score by Parental Education chart shows 71 for the selected “Some College” category.

Interpretation: This value describes the currently displayed category. Other categories should be compared under consistent filter conditions before drawing a broader conclusion.

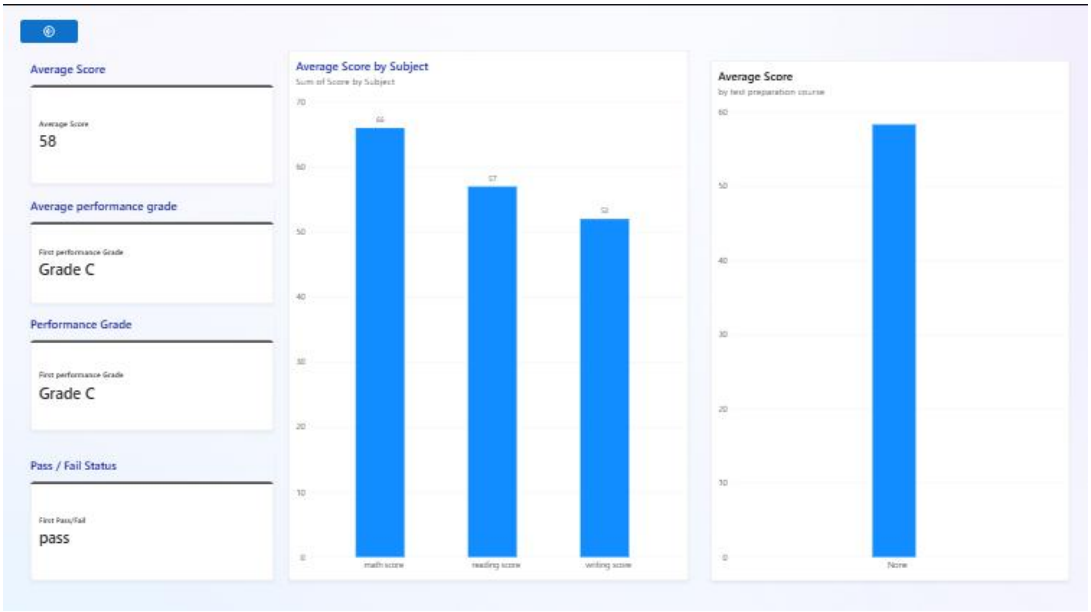
The dashboard values change dynamically based on the slicer and filter selections. Therefore, the observations presented in this report are based on the default dashboard view.

7. FINAL DASHBOARD VALIDATION

The dashboard was reviewed using the following checks:

- Slicers were checked for correct filtering.
- Visual interactions were reviewed.
- Drill-through navigation to the Academic Profile page was tested.
- The Back button was tested successfully.
- Measures, titles, and labels were reviewed.
- Visuals were checked for blank or error displays.
- The dashboard was reviewed to ensure that it supports the intended academic analysis.





8. PROJECT OUTPUT

The final output is an interactive Student Performance Analytics Dashboard containing KPI cards, subject comparisons, demographic analysis, test preparation analysis, grade distribution, slicers, and an Academic Profile drill-through page.

The dashboard provides a visual summary of student performance and supports further exploration of the dataset.

9. CONCLUSION

The Student Performance Analytics Dashboard was developed using Microsoft Power BI to analyse academic scores and related student information. The dashboard helps compare subject-wise performance, examine test preparation groups, explore demographic categories, and review grade distribution.

The drill-through page provides a detailed Academic Profile, while the Back button enables navigation to the main dashboard. The project demonstrates how interactive visualizations can be used to summarize data and identify evidence-based performance patterns.